

PowerFlow: Unlocking the Dual Nature of LLMs via Principled Distribution Matching

One tunable knob α — sharpen to reason, flatten to create.

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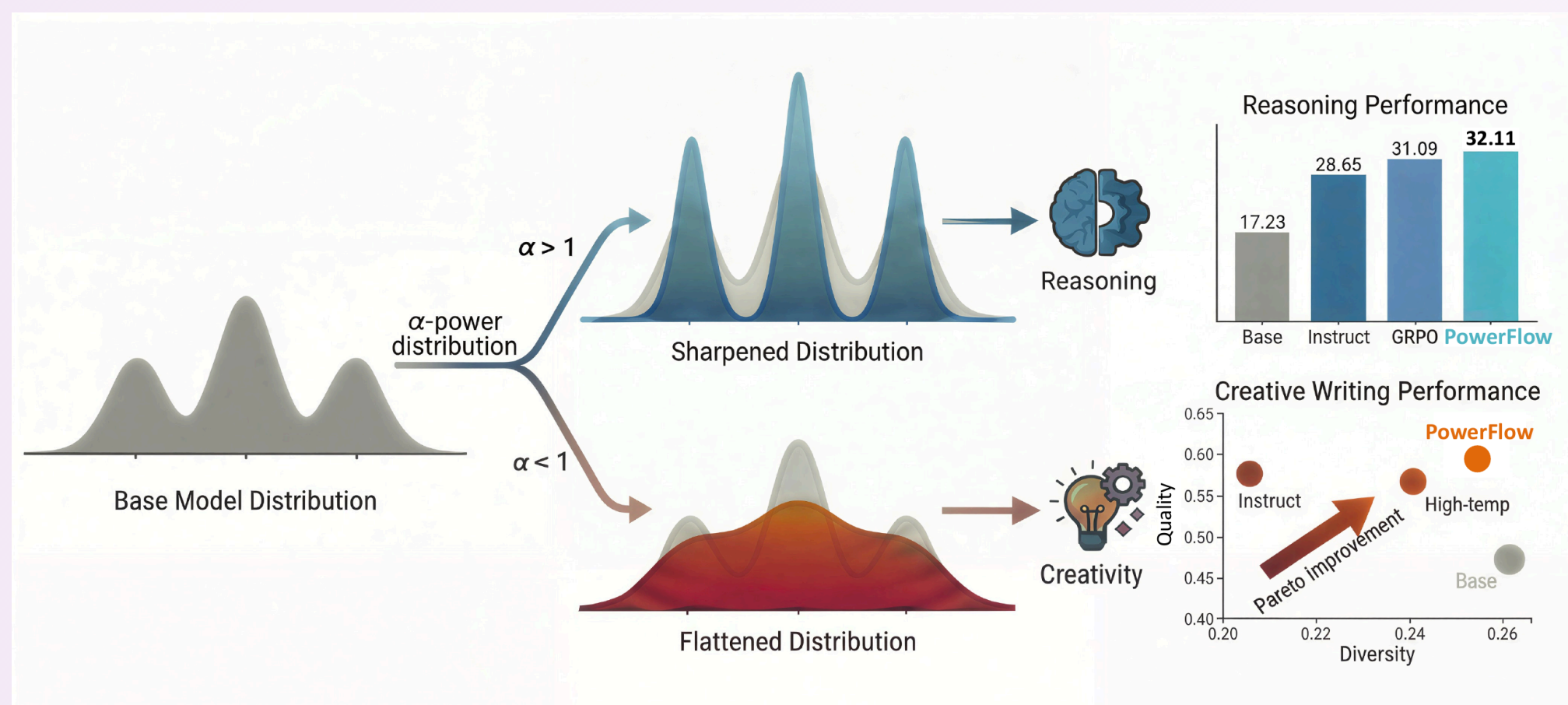
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Code & Models



FRAMEWORK PowerFlow matches a length-aware α -power target of the base model. A single knob α directionally elicits **reasoning** ($\alpha > 1$, sharpen) or **creativity** ($\alpha < 1$, flatten) — fully unsupervised under standard decoding.

+1.7

avg@16 over GRPO
(Qwen2.5-1.5B)

Stable

monotonic gains;
no length collapse

Pareto

quality $\times$ diversity
across 4 models

I-proj.

minimum-KL
length correction

1 Motivation

Unsupervised **RLIF** elicits latent LLM capabilities, but its *heuristic* rewards cause: **length collapse**, **overconfidence**, **mode collapse**, & reward hacking.

Q: Replace handcrafted rewards with a principled, controllable target distribution?

2 Core Idea: α -Power

Recast fine-tuning as **distribution matching**. Target the α -power (escort) distribution of the base model — entropy modulates while **rankings & mode structure are strictly preserved**:

TARGET DISTRIBUTION

$$p_{\alpha}(y | q) \propto p_{\text{base}}(y | q)^{\alpha}$$

$\alpha > 1$

Sharpen · Reason

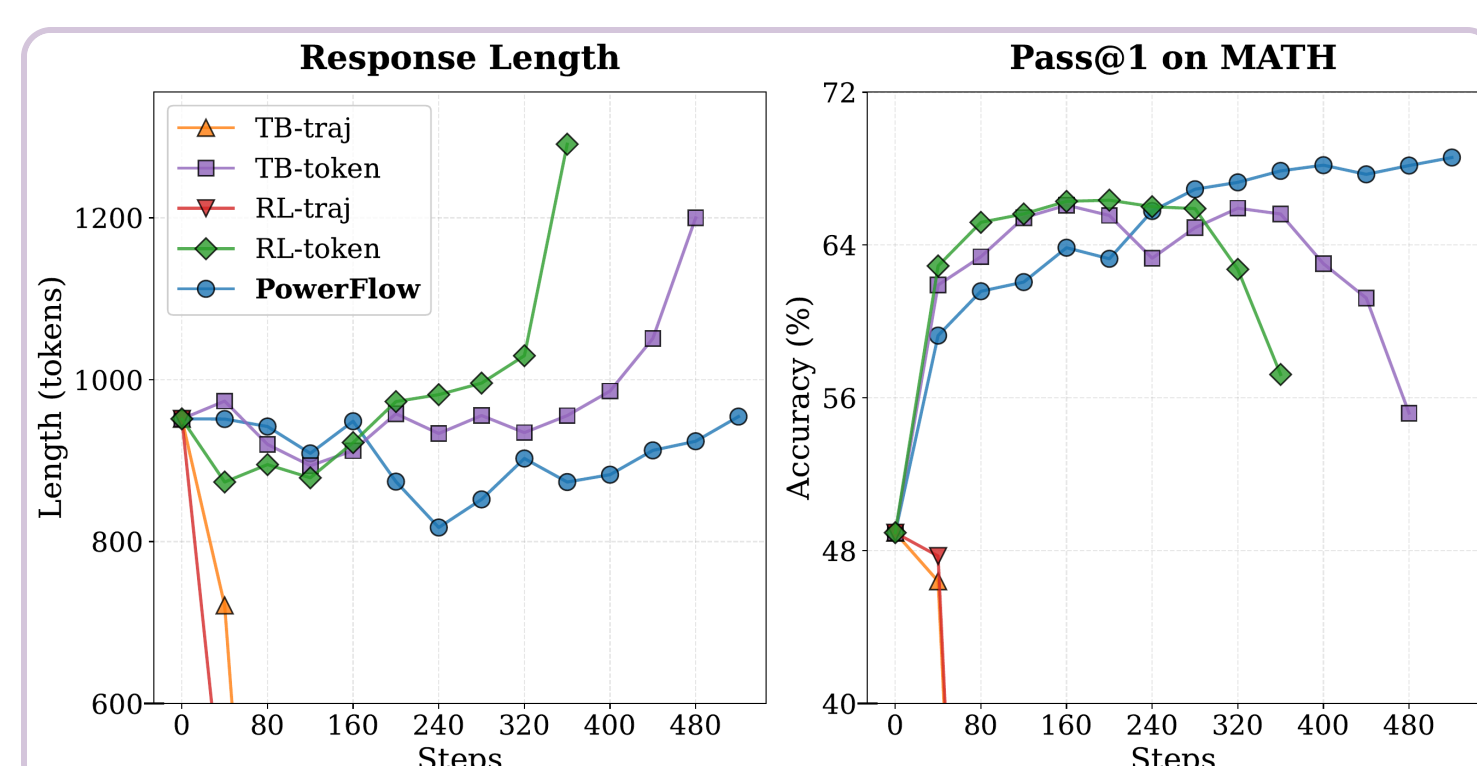
$\alpha = 1$

Base · Identity

$\alpha < 1$

Flatten · Create

3 Why Length-Aware? ★ Evidence



Naïve TB / RL:

length collapse at $\alpha > 1$.

Token TB: repetition decay at $\alpha < 1$.

PowerFlow stays length-stable with monotonic gains on MATH.

4 GFlowNet as Amortized Sampler

Bypass the intractable $Z(q)$ via **Trajectory Balance** (TB), a variational surrogate that at equilibrium of Z_{ϕ} satisfies $\mathbb{E}[\nabla_{\theta} \mathcal{L}_{\text{TB}}] = 2\nabla_{\theta} \mathbb{D}_{\text{KL}}(P_F \| p_{\text{target}})$.

STANDARD TB (AUTOREGRESSIVE)

$$\mathcal{L}_{\text{TB}} = \left(\log Z_{\phi} + \sum_{t=1}^T \log \pi_{\theta}(y_t | y_{<t}, q) - \log \tilde{p}_{\text{target}} \right)^2$$

But: on LLMs this is **dominated by sequence length**, not semantic density — causing *length collapse* ($\alpha > 1$) or *repetition explosion* ($\alpha < 1$).

5 Length-Aware TB ★ KEY

Reparameterize $Z_{\phi}(q, y) = (Z'_{\phi}(q))^{|y|}$ and normalize by $|y|$ — projecting onto a length-normalized energy surface:

POWERFLOW LOSS (w : CLIPPED IS; ψ : FORMAT PENALTY)

$$\mathcal{L}_{\text{PF}} = w \left(\log Z'_{\phi} + \frac{1}{|y|} \log \pi_{\theta} - \alpha \left[\frac{1}{|y|} \log p_{\text{base}} + \psi \right] \right)^2$$

The induced target is a 1-D exponential tilt of p_{α} by $|y|$: $\pi^* \propto p_{\text{base}}^{\alpha} e^{-\lambda_q |y|}$.

Theory. LA-TB is the *I-projection* — the minimum-KL length correction onto length-calibrated distributions.

Empirical: pairwise inversion rate vs. the ideal α -power target on Qwen2.5-Math-1.5B is **IR $\approx$ 0.09** — ~91% of rankings preserved.

6 Reasoning ($\alpha > 1$) ★ GRPO-Matched

Avg@16 on MATH500 · Olympiad · AIME24/25 · AMC23 · GPQA

Method	Avg.	Δ GRPO
Qwen2.5-1.5B		
Intuitior	18.95	+0.82
PowerFlow	19.85	+1.72
GRPO (sup.)	18.13	ref
Qwen2.5-Math-1.5B		
EMPO	32.45	−0.30
PowerFlow	34.30	+1.55
GRPO (sup.)	32.75	ref
Llama-3.2-3B-Inst.		
PowerFlow	22.88	+0.55
GRPO (sup.)	22.33	ref
Qwen2.5-Math-7B		
EMPO / TTTL	40.88 / 41.18	−1.50 / −1.20
PowerFlow	42.17	−0.21
GRPO (sup.)	42.38	ref

7 Reasoning Diversity Preserved

4.05

PowerFlow

3.93

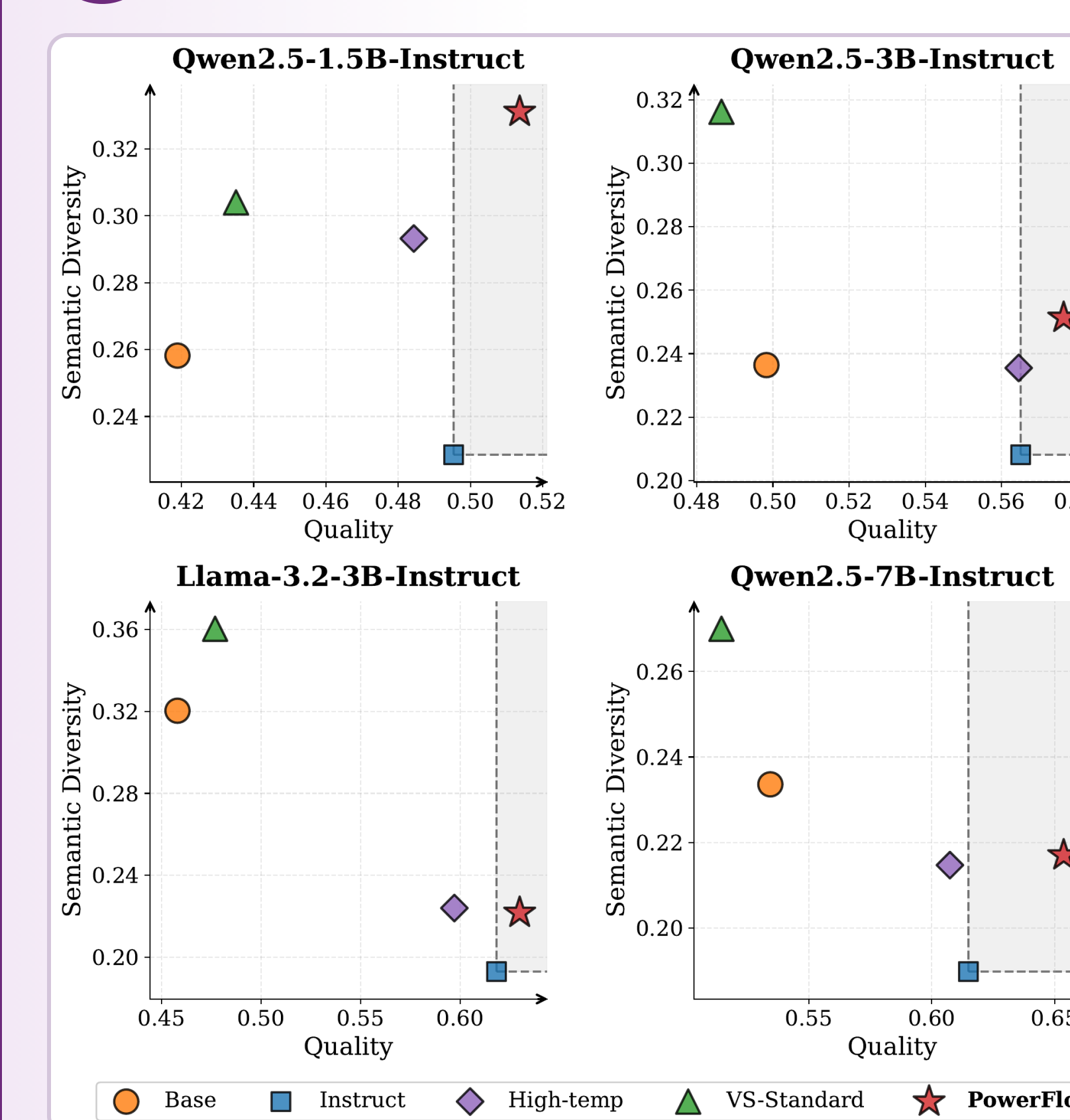
GRPO (sup.)

3.80

EMPO (RLIF)

Strategy-diversity on AIME24/25 (DeepSeek-V3.2 judge, 16 samples/problem). The α -power transform preserves the base model's multi-modal structure — **no monolithic mode collapse**.

8 Creativity ($\alpha < 1$) ★ Pareto-Dominant



AXES

Quality $\times$ Diversity

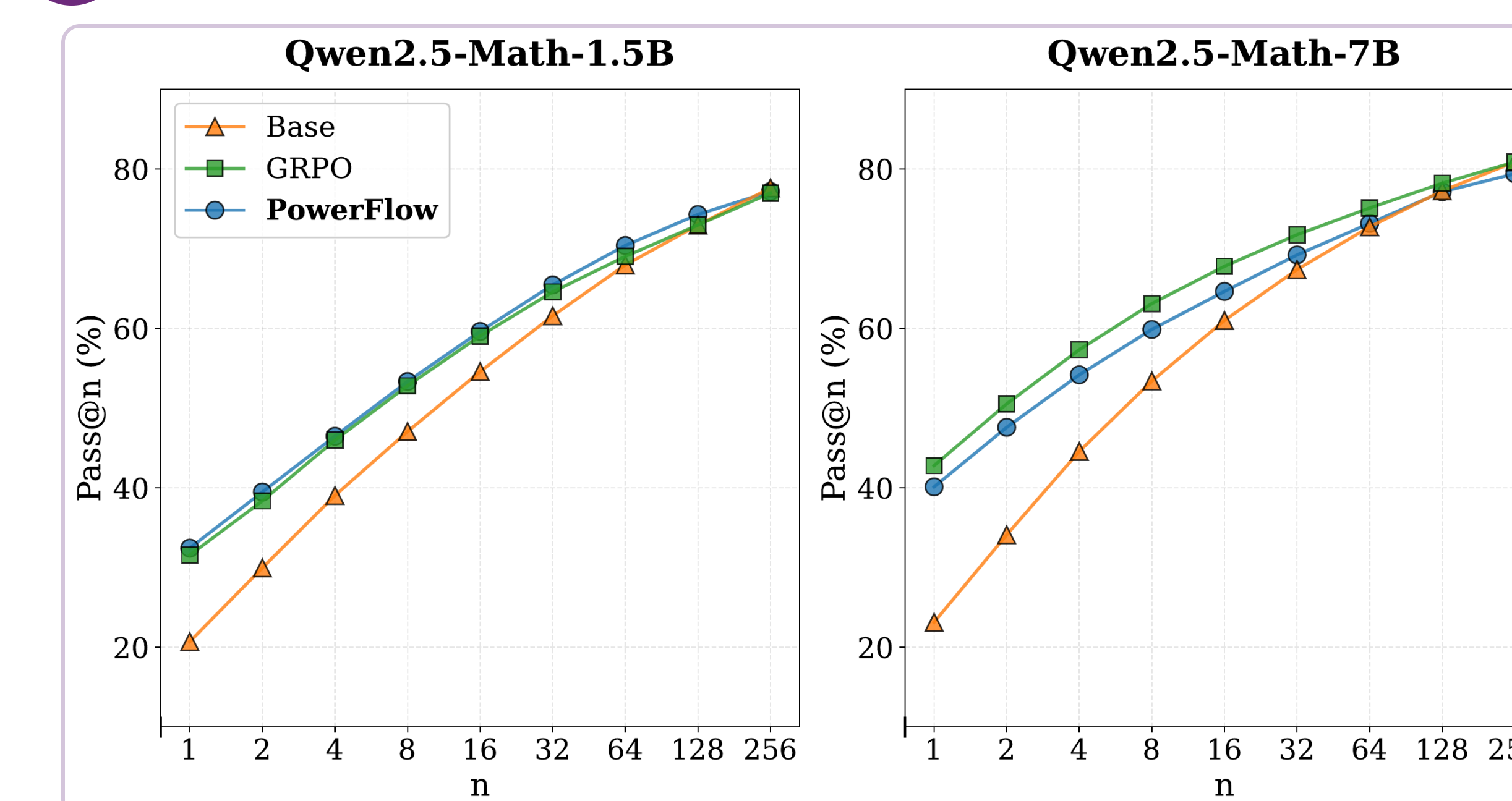
$\nearrow$ upper-right is better

★ **RESULT**
Beyond the Instruct frontier

SETUP
4 model families

$\alpha = 0.5$ flatten

9 Reasoning Mechanism: Pass@n



PowerFlow tracks GRPO across $n = 1 \rightarrow 256$ on **OlympiadBench** — sharpening lifts pass@1 *without* sacrificing reasoning coverage.

10 Takeaways

PRINCIPLED. One knob α replaces all heuristic rewards.

LENGTH-AWARE. I-projection neutralizes length bias.

DUAL-NATURE. Reasoning & creativity, one loss.

PRACTICAL. ~10% step overhead; standard decoding.